



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

RESOURCEHUB: A SECURE CAMPUS RESOURCE SHARING AND EXCHANGE PLATFORM WITH QR-BASED HANDOVER VERIFICATION

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the course of

CSA4305 – INTERNET PROGRAMMING

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted By

Deepak V (192411021),

Under the Supervision of

DR. D. Sudhagar

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

SEPTEMBER 2026



**SIMATS ENGINEERING,
SAVEETHA INSTITUTE OF
MEDICAL AND
TECHNICAL SCIENCE, CHENNAI-
602105**

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "Campus Resource Exchange Network(CREN): An Intelligent Academic Marketplace for Resource Sharing" has been carried out by Deepak V (192411021), under the supervision of DR. D. Sudhagar and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR	COURSE FACULTY
Dr Marie Claude .N	Dr.D.Sudhagar
Professor	Professor
Department of Neural Networks	Quantum Intelligence
SIMATS Engineering,	SIMATS Engineering,
SIMATS, Chennai – 602105.	SIMATS, Chennai – 602105.

Submitted for the Project work Viva-Voce held on _____.

**INTERNAL EXAMINER
EXAMINER**

EXTERNAL

DECLARATION

I, Deepak V (192411021) of the CSE Department, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Campus Resource Exchange Network(CREN): An Intelligent Academic Marketplace for Resource Sharing" is the result of my own bonafide efforts. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. I confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:

Date:

Signature of Student

Deepak V (192411021),

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCE

CHENNAI- 602105

Individual Contribution Statement

Student Name / Register No.	Module / Responsibility	Design & Development	Testing & Analysis	Report Contribution	Contribution
Meghanadh A (192372281)	Module 1 – User Authentication	Authentication & user management	Authentication testing	Module 1 documentation	20%
Aaron Samuel S (192411022)	Module 2 – Resource Management	Resource CRUD & categories	Resource testing	Module 2 documentation	20%
Deepak V (192411021)	Module 3 – Exchange Management & Trust System	Exchange requests & transactions, QR verification & trust logic	Transaction & trust-score testing	Module 3 documentation	20%
Y. Abhay (192411219)	Module 4 – QR Verification & Trust System	QR verification & trust system	QR & trust testing	Module 4 documentation	20%
Ebin Antony P (192373011)	Module 5 – Supporting Features	Notifications, wishlist & reports	Feature testing	Module 5 documentation	20%
Total					100%

ABSTRACT

ResourceHub is a student-focused academic resource exchange platform designed to provide a secure and reliable environment for students to exchange books, electronics, laboratory equipment, project components, stationery, and other academic resources. The system combines user authentication, resource management, exchange requests, transaction management, QR-based handover verification, and a peer-based trust system. This report focuses on Module 3: Exchange Management & Trust System, which is responsible for securely handling the exchange of academic resources between students and for maintaining a reliable reputation mechanism across the platform.

The Exchange Management component processes every exchange request raised by a student who wishes to buy, sell, lend, donate, or swap a resource. Once a student selects a resource, the system allows the request to be sent, accepted, rejected, or completed, and a corresponding exchange transaction is created the moment a request is accepted. Each transaction record stores complete information about the exchange, including the participating users, the resource involved, the relevant dates, and the current transaction status, so that the full lifecycle of an exchange can always be traced.

QR-code verification is used as the mechanism that confirms the physical handover between the buyer and the seller. A unique QR code is generated for every exchange transaction, and both participants must complete the verification step before the transaction can move to a completed state. This prevents an exchange from being marked as finished without an actual handover taking place. Once an exchange is completed, participants can rate and review each other; these ratings and reviews are combined with prior transaction history to calculate and continuously update each student's trust score.

Overall, Module 3 improves the reliability of the Campus Resource Exchange Network (CREN) by ensuring that exchanges are verified and properly recorded. It supports the project's broader goal of creating a trusted academic resource-sharing environment by tying together exchange-request processing, QR-based verification, transaction record-keeping, and trust-score calculation into a single, dependable subsystem.

Keywords: Exchange Management, Trust System, QR Code Verification, Exchange Transaction, Transaction Records, Ratings and Reviews, Trust Score, Resource Exchange, Campus Resource Exchange Network, ResourceHub.

TABLE OF CONTENTS

S.NO	CONTENTS	PG.NO
1.	Introduction	9-10
	1.1 Background Information	9
	1.2 Project Objectives	9
	1.3 Significance	9
	1.4 Scope	10
	1.5 Methodology Overview	10
2.	Problem Identification and Analysis	11-12
	2.1 Description of the problem	11
	2.2 Evidence of the problem	11
	2.3 Stakeholders	11
	2.4 Supporting Data/Research	12
3.	Solution Design and Implementation	13-15
	3.1 Development & Design process	13
	3.2 Tools & Technologies used	13
	3.3 Solution Overview	14
	3.4 Engineering Standards Applied	14
	3.5 Solution Justification	15
4.	Results and Recommendations	16-18
	4.1 Screenshots	16
	4.2 Evaluation of Results	17
	4.3 Challenges Faced	17
	4.4 Proposed Improvements	18
	4.5 Future Recommendations	18
5.	Reflection on Learning and Personal Development	19-21
	5.1 Key Learning Outcomes	19
	5.2 Challenges Encountered and Overcome	19
	5.3 Applications of Engineering Standards	20
	5.4 Industry Insights	20
	5.5 Conclusion of Personal Development	21
6.	Conclusion	22

7.	References	23
8.	Appendices	24-29

LIST OF FIGURES

FIG NO	Figure Name	PAGE NO
3.1	Exchange Request Interface (Send / Accept / Reject Request)	16
3.2	Exchange Transaction Created & QR Code Generated	16
3.3	QR Code Verification Screen (Buyer–Seller Handover)	17
3.4	Rating & Review Submission Interface	17
3.5	Trust Score Display & Transaction History	17

LIST OF TABLES

TABLE NO	Table Name	PAGE NO
3.2	Tools & Technologies Used	13
4.2	Evaluation of Results	17
E.1	API Endpoints	26
G.1	Testing Checklist	29

LIST OF ABBREVIATIONS / SYMBOLS

Symbol / Abbreviation	Description	Unit
CREN	Campus Resource Exchange Network	—
QR	Quick Response Code	—
JWT	JSON Web Token	—
API	Application Programming Interface	—
REST	Representational State Transfer	—
DB	Database	—
ID	Identifier	—
HTTP	Hypertext Transfer Protocol	—
UI	User Interface	—
URL	Uniform Resource Locator	—
CRUD	Create, Read, Update, Delete	—
E2E	End-to-End (testing)	—

CHAPTER 1

INTRODUCTION

1.1 Background Information

Students frequently need to buy, sell, lend, donate, or swap academic resources such as textbooks, laboratory equipment, electronic components, project materials, and stationery. When this is handled informally, it becomes difficult to keep track of who currently holds a resource, whether an agreed exchange actually took place, and whether the other participant can be trusted for a future exchange.

The Exchange Management & Trust System addresses this gap within ResourceHub. When a student wants to exchange a resource, the module processes the request end-to-end — from the initial request, through acceptance, to a verified handover — and keeps a permanent record of the transaction. QR-code verification is used to confirm that the handover between the buyer and the seller actually happened, while the accumulated transaction history, ratings, and reviews are used to calculate and continuously update each student's trust score.

By combining structured exchange-request processing with QR-based verification and a peer-review-driven trust score, Module 3 gives ResourceHub a dependable backbone for every exchange that takes place on the platform.

1.2 Project Objectives

The main objective of Module 3 is to provide a secure, traceable, and trustworthy mechanism for exchanging academic resources between students.

The specific objectives are:

- To allow students to send exchange requests for resources they wish to buy, sell, lend, donate, or swap.
- To allow the resource owner to accept or reject an incoming exchange request.
- To automatically create an exchange transaction once a request is accepted.
- To generate a unique QR code for every exchange transaction.
- To use QR-code scanning to verify that the physical handover between buyer and seller has taken place.
- To prevent an exchange from being marked completed without successful QR verification.

- To store complete transaction records, including users, resource, dates, and status.
- To allow students to rate and review each other after a completed exchange.
- To calculate a trust score for each user based on transaction history and interactions.
- To update a user's trust score whenever a new rating is submitted.
- To display trust scores and review information so future users can identify reliable students.
- To improve the overall reliability of the Campus Resource Exchange Network (CREN).

1.3 Significance

The Exchange Management & Trust System is significant because it addresses two challenges that are central to any peer-to-peer exchange platform: coordinating the exchange itself, and helping students decide whether a fellow participant can be trusted.

Structured exchange-request processing ensures that a resource cannot simply change hands without a clear request, acceptance, and completion sequence. This gives both participants a shared, auditable record of what was agreed and when it took place.

QR-code verification adds a physical-world confirmation step before a transaction is allowed to reach the completed state. This reduces the possibility of an exchange being recorded as successful when no handover actually occurred, and it gives both students confidence that the system reflects what really happened.

The trust system is significant because reputation is one of the strongest signals available in a peer-to-peer marketplace. A student who has completed several exchanges successfully and received positive ratings builds a visible, quantifiable reputation, while a pattern of poor ratings warns future users to be cautious. Together, verified exchanges and trust scoring make CREN considerably more reliable than a simple listing-and-messaging system.

1.4 Scope

The scope of Module 3 covers exchange-request processing, transaction creation, QR-code generation and verification, transaction record-keeping, and the ratings, reviews, and trust-score mechanism within ResourceHub.

The module includes:

- Sending, accepting, rejecting, and completing exchange requests.

- Support for buy, sell, lend, donate, and swap exchange types.
- Automatic creation of an exchange transaction on request acceptance.
- Generation of a unique QR code per transaction.
- QR-code scanning and verification between buyer and seller.
- Preventing transaction completion without QR verification.
- Storage of complete transaction records (users, resource, dates, status).
- Submission of post-exchange ratings and written reviews.
- Trust-score calculation based on transaction history and ratings.
- Trust-score and review display for future users.
- End-to-end testing of exchange, verification, and trust workflows.

The module does not replace the authentication or resource-listing components of ResourceHub. Instead, it works together with those modules — relying on authenticated users and listed resources — to provide the exchange and trust functionality described above.

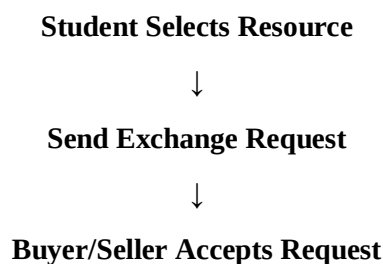
1.5 Methodology Overview

The implementation of Module 3 followed a structured software development approach consisting of requirement identification, database design, backend implementation, QR-verification integration, trust-score logic, frontend integration, and end-to-end testing.

The exchange process begins when a student selects a resource and sends an exchange request to the resource owner. The buyer or seller can then accept or reject the request. When a request is accepted, an exchange transaction record is created and a unique QR code is generated for that transaction. During the physical handover, the QR code is scanned and verified by the counterparty. Once verification succeeds, the exchange is marked completed, after which both participants can leave a rating and review. Each new rating feeds into the trust-score calculation, and the finalised transaction record is stored permanently.

The overall module workflow can be represented as follows:

Module Workflow:





Exchange Transaction Created



QR Code Generated



QR Code Verification



Exchange Completed



Rating & Review



Trust Score Updated



Transaction Record Stored

CHAPTER 2

PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Peer-to-peer resource exchange on a student marketplace involves more than simply matching a buyer with a seller. An exchange request being accepted does not, by itself, guarantee that the resource actually changed hands, and it says nothing about whether either participant can be relied upon in future exchanges.

If a system allows an accepted request to be marked as completed without any further confirmation, a student could report an exchange as finished even though no physical handover occurred, leading to disputes and inaccurate records. Without a verification step tied specifically to the two participants and the specific transaction, there is also no reliable way to confirm that it was genuinely the intended buyer and seller who met.

A further problem is the absence of trust information. A resource listing alone tells a student nothing about whether the owner has completed past exchanges reliably, responds to requests, or has received negative feedback from previous participants. Without a trust mechanism, students have no way to judge risk before agreeing to an exchange.

Therefore, Module 3 needs to:

- Provide a clear, auditable exchange-request workflow (send, accept, reject, complete).
- Create a transaction record the moment a request is accepted.
- Verify that a physical handover has actually taken place before completion.
- Tie verification to the specific transaction and its two participants.
- Store complete and accurate transaction history.
- Allow participants to rate and review each other after completion.
- Calculate and maintain a meaningful trust score for every user.

2.2 Evidence of the Problem

The need for a controlled exchange workflow can be seen in the lifecycle of a typical transaction. Once a request is accepted, the resource is effectively reserved for the requester, but the system still needs proof that the two students actually met and exchanged the item before the transaction can be closed.

Without QR-based verification, a transaction could move directly from accepted to completed on the basis of a single click, with no independent confirmation from the other participant. Module 3 avoids this by requiring a successful QR-code scan from both sides before completion is permitted.

The absence of a trust mechanism is also evident from user feedback during design discussions: students repeatedly asked how they would know whether an unfamiliar classmate listing a resource was reliable. This confirmed the need for a rating-and-review system that feeds into a visible trust score.

2.3 Stakeholders

Students / Resource Owners

Resource owners receive and respond to exchange requests, participate in QR verification during handover, and benefit from a documented transaction history and trust score.

Students / Requesters

Requesters send exchange requests, scan the QR code during handover, and can rate and review the resource owner after a completed exchange.

Exchange Participants

Both sides of an exchange contribute to and rely on the trust system through completed transactions, ratings, and reviews.

System Administrators

Administrators benefit from accurate, well-structured transaction records when monitoring platform activity.

ResourceHub Developers

Developers are responsible for implementing secure exchange-request handling, QR verification, transaction storage, and trust-score logic.

Academic Institution

The institution benefits from a more structured, accountable, and trustworthy environment for students to exchange academic resources.

2.4 Supporting Data/Research

Transaction-based workflows are widely used in marketplace and e-commerce systems to ensure that every exchange passes through clearly defined states — such as requested,

accepted, completed, or cancelled — rather than being represented by a single ambiguous flag. Module 3 follows this pattern for exchange requests and transactions.

QR codes are a well-established mechanism for linking a physical action to a digital record, since they can encode a unique, transaction-specific value that is quick to generate and scan. Module 3 applies this principle by generating a unique QR code per transaction and requiring it to be verified before completion.

Reputation systems based on peer ratings are a common feature of trusted peer-to-peer platforms, because they give participants a quick, quantifiable signal about another user's reliability. Module 3 applies this principle to campus resource sharing by combining transaction history with ratings and reviews to calculate a trust score for every student.

These principles — structured transaction states, QR-based physical confirmation, and peer-review-driven reputation — provide the technical foundation for the Exchange Management & Trust System implemented in this module.

CHAPTER 3

SOLUTION DESIGN AND IMPLEMENTATION

The development of Module 3 followed several stages.

Requirement Identification

The first stage involved identifying the need for a structured exchange-request workflow, a physical-handover verification mechanism, and a trust-scoring system for ResourceHub.

Exchange Request Design

A request-based workflow was designed to let a student select a resource and send an exchange request, which the resource owner could then accept, reject, or leave pending.

Transaction Creation

Accepting a request automatically creates an exchange transaction record, storing the participating users, the resource, and the relevant dates, with its status initialised for the upcoming handover.

QR Verification Design

A unique QR code is generated for each transaction. Both the buyer and the seller are required to complete a QR-code verification step during the physical handover before the transaction can be marked completed.

Trust Score Design

A trust-score calculation was designed to combine a user's completed-transaction history with the ratings and reviews received from other students, producing a single reputation value that is updated after every new rating.

Ratings and Reviews Implementation

Once an exchange is completed, both participants are permitted to submit a rating and an optional written review for one another, which are stored and linked to the specific transaction.

Frontend Integration

Interactive exchange-request screens, QR-code display and scanning, rating and review submission, and trust-score display were integrated into the ResourceHub frontend.

Testing and Validation

End-to-end tests were performed to validate exchange-request handling, transaction creation, QR generation and verification, trust-score updates, and transaction-record accuracy.

3.2 Tools & Technologies Used

Technology	Purpose
Node.js	Backend runtime environment
Express.js	REST API and backend routing
MySQL	Storage of users, resources, exchange requests, transactions, and reviews
mysql2	MySQL database connectivity
JWT	Authentication and user identity verification
React	Frontend user interface
Vite	Frontend development and production build
Axios	Frontend-to-backend API communication
qrcode.react	QR code generation in the frontend
html5-qrcode	Camera-based QR code scanning
REST API	Communication between frontend and backend modules
MySQL Transactions	Atomic exchange and transaction state updates
PowerShell	API and backend testing
E2E Test Scripts	Automated validation of exchange and trust workflows

These technologies work together to implement the exchange-management and peer-based trust functionality of ResourceHub.

3.3 Solution Overview

Module 3 consists of three primary components: Exchange Request Processing, QR-Based Handover Verification, and the Trust System.

Exchange Request Processing

A student selects a resource and sends an exchange request, specifying whether it is a buy, sell, lend, donate, or swap. The resource owner can accept, reject, or leave the request pending. When a request is accepted, the resource is reserved and an exchange transaction is created.

The transaction record stores:

- Transaction ID
- Buyer and seller identifiers
- Resource ID

- Exchange type
- Transaction status
- Creation timestamp
- Completion timestamp

QR-Based Handover Verification

Once a transaction is created, a unique QR code is generated for it. During the physical handover, the QR code is scanned and verified between the buyer and the seller. Only after successful verification can the transaction move to the completed state, ensuring that a transaction cannot be closed without confirmation from both participants.

Trust System

After a transaction is completed, both participants can submit a rating (typically on a one-to-five scale) and an optional written review for one another. The review contains:

- Transaction ID
- Reviewer ID
- Reviewed user ID
- Rating value
- Optional review text
- Creation timestamp

The resulting reviews, combined with a user's transaction history, are used to calculate and continuously update the trust score, which is then displayed to future students evaluating whether to exchange with that user.

Combined Module Workflow:

Student Selects Resource → Send Exchange Request → Request Accepted → Exchange Transaction Created → QR Code Generated → QR Code Verified → Exchange Completed → Rating & Review → Trust Score Updated → Transaction Record Stored

3.4 Engineering Standards Applied

Authentication and Authorization

Authenticated users are identified using JWT-based authentication, and backend checks ensure that only the correct buyer or seller can act on a given exchange request or transaction.

Least Privilege

Only the resource owner can accept or reject a request for their own resource, and only the two participants in a transaction can perform QR verification and submit reviews for that transaction.

Defense in Depth

Multiple controls — authentication, authorization, transaction-state checks, QR verification, and duplicate-review prevention — work together instead of relying on a single safeguard.

Input Validation

The backend validates request data, transaction identifiers, QR tokens, and rating values before modifying the database.

Database Transaction Integrity

Critical operations such as transaction creation and status updates use database transactions to maintain consistent state.

Duplicate Prevention

Backend validation prevents duplicate reviews for the same transaction and reviewer.

Backend Enforcement

Business rules such as "no completion without QR verification" are enforced on the backend rather than relying solely on the frontend.

Auditability and Traceability

Every transaction and review record stores the relevant user and resource information, allowing exchanges to be traced and audited later.

3.5 Solution Justification

A request-based workflow (send, accept, reject, complete) was chosen instead of an instant exchange so that both participants have a clear, mutually agreed record of what was requested and when it was accepted.

QR codes were selected as the handover-confirmation mechanism because they provide a fast, low-friction way for two students to confirm a physical exchange using devices they already carry — the seller can display the code while the buyer scans it, or vice versa.

Requiring QR verification before completion was chosen because verification is only meaningful if it directly affects the transaction lifecycle; a transaction that can be completed without any confirmation would defeat the purpose of adding QR verification at all.

The trust system was implemented using peer ratings and reviews because the two actual participants in an exchange are best positioned to judge the reliability of the process and each other. Combining ratings with completed-transaction history, rather than ratings alone, produces a more stable and meaningful trust score.

Overall, the combination of structured exchange-request handling, QR-based verification, and trust scoring gives ResourceHub both transaction-level assurance and user-level reputation, making the platform considerably more reliable than a simple listing-and-messaging system.

CHAPTER 4

RESULTS AND RECOMMENDATIONS

4.1 Screenshots

The following illustrative screens represent the intended Exchange Management & Trust System interfaces; they should be replaced with actual application screenshots before final submission.



Figure 3.1: Exchange Request Interface (Send / Accept / Reject Request)

This screen represents the interface used by a student to select a resource and send an exchange request, and the interface used by the resource owner to accept or reject an incoming request.

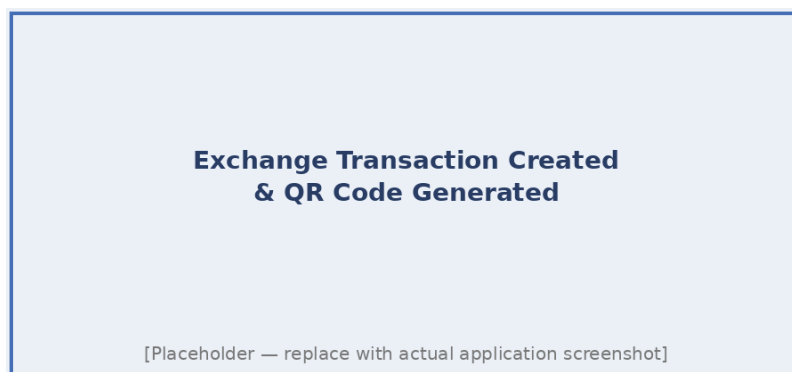


Figure 3.2: Exchange Transaction Created & QR Code Generated

This screen represents the newly created exchange transaction along with the unique QR code generated for the handover.



Figure 3.3: QR Code Verification Screen (Buyer–Seller Handover)

This screen represents the QR-code scanning and verification step performed by the participants during the physical handover.



Figure 3.4: Rating & Review Submission Interface

This screen represents the rating-and-review interface shown to participants after a transaction is completed.

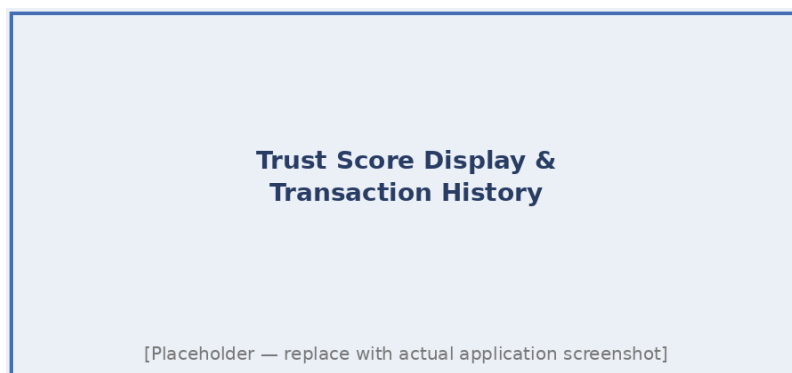


Figure 3.5: Trust Score Display & Transaction History

This screen represents a student's trust score, alongside their past exchange transaction history.

4.2 Evaluation of Results

The implementation of Module 3 was evaluated through functional, database, and end-to-end test cases.

Test Case	Expected Result
Send exchange request for available resource	Request is created successfully
Owner accepts exchange request	Transaction is created and resource reserved
Owner rejects exchange request	Request status updated, resource remains available
Generate QR code for accepted transaction	Unique QR code generated and linked to transaction
Complete transaction before QR verification	Completion is rejected
Verify QR code between correct participants	Transaction proceeds to completed state
Submit rating after completed exchange	Rating stored and trust score updated
Submit duplicate review for same transaction	Request is rejected
Retrieve transaction history	Complete and accurate records returned
Trust score regression test	All exchange and trust E2E checks pass

The evaluation demonstrates that Module 3 successfully processes exchange requests, creates accurate transaction records, enforces QR verification before completion, and maintains a functioning trust-scoring mechanism.

4.3 Challenges Faced

One challenge during implementation was ensuring that a transaction could only be created once a request had genuinely been accepted, and that the resource's availability status was updated consistently alongside it.

Another challenge was tying QR verification correctly to a specific transaction and its two participants, so that neither an unrelated student nor a repeated scan of an already-verified code could affect the transaction state.

Calculating the trust score in a way that fairly reflected both transaction history and incoming ratings, without letting a single rating swing the score too heavily, required iterating on the calculation logic.

Preventing duplicate reviews for the same transaction and reviewer also required careful backend validation, since a naive implementation allowed a participant to submit multiple ratings for the same exchange.

These challenges improved practical understanding of transaction-state management, verification design, and reputation-scoring logic.

4.4 Proposed Improvements

The exchange-request interface could provide clearer real-time status updates when a request is accepted or rejected, rather than requiring the student to refresh the page.

Rate limiting could be introduced on QR-verification attempts to reduce the risk of repeated or automated verification requests.

The trust-score calculation could be extended to weight more recent transactions more heavily, so that a student's current reliability is better reflected than very old activity.

Additional moderation tools could allow administrators to review flagged or suspicious ratings and reviews.

4.5 Future Recommendations

Future versions of Module 3 could support push notifications when a request is accepted, when a QR code is generated, and when a review is received.

The trust-score model could be expanded into a more comprehensive reputation system that also considers cancellation history, response time, and repeat-exchange frequency.

The QR-verification mechanism could be extended with dynamically refreshed codes and stronger anti-replay protection for higher-value exchanges.

CHAPTER 5

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

Working on Module 3 provided practical experience in designing a multi-stage transaction workflow, from an initial request through acceptance, verification, and completion.

I learned how QR codes can be used as a bridge between a physical action and a digital transaction record, and how to design a verification step that meaningfully gates a transaction's progress rather than being a cosmetic addition.

Implementing the trust-score logic improved my understanding of how reputation systems combine multiple signals — transaction history, ratings, and reviews — into a single, meaningful score.

The module also improved my understanding of relational database design, since exchange requests, transactions, and reviews all needed to be linked correctly to users and resources while keeping the data consistent.

5.2 Challenges Encountered and Overcome

During development, one challenge was correctly sequencing the transaction states — request, accepted, QR-verified, completed — so that a transaction could not skip a required step. This was resolved by adding explicit backend checks at each transition.

Another challenge involved coordinating the QR-generation and QR-scanning views so that each participant saw the correct interface for their role in the transaction, which required carefully separating buyer-side and seller-side logic.

Designing the trust-score calculation also required experimentation, since an overly simple average of ratings did not account for transaction volume, while a more complex formula risked being difficult to explain to users.

5.3 Applications of Engineering Standards

The principle of least privilege was applied by restricting exchange-request actions and QR verification to the correct participants for each transaction.

Defense in depth was achieved by combining authentication, authorization checks, transaction-state validation, and QR verification rather than relying on any single control.

Database integrity principles were applied through foreign-key relationships between requests, transactions, users, and resources, along with transaction-based updates for critical state changes.

Duplicate prevention was implemented for reviews so that the same reviewer cannot submit multiple ratings for the same transaction.

5.4 Industry Insights

Developing this module highlighted that authentication alone is not sufficient for a marketplace feature — authorization checks specific to a transaction's participants are equally important.

I also learned that physical-confirmation mechanisms such as QR codes are a practical and widely applicable way to connect real-world actions to digital records, useful well beyond this specific project.

The trust-system work demonstrated how reputation mechanisms shape user behaviour in peer-to-peer platforms, encouraging participants to follow through on agreed exchanges.

5.5 Conclusion of Personal Development

Working on Module 3 significantly improved my understanding of transaction-oriented system design, verification mechanisms, and reputation scoring.

The experience strengthened my skills in backend API design, database schema design, and coordinating a multi-step workflow across both the frontend and backend of a real application.

CHAPTER 6

CONCLUSION

Module 3, Exchange Management & Trust System, provides the core transaction backbone of the ResourceHub platform. It securely handles the exchange of academic resources between students, from the initial request through acceptance, QR-based verification, and completion.

The exchange-request component ensures that every buy, sell, lend, donate, or swap request follows a clear, auditable workflow, and that an exchange transaction is created and correctly tracked from the moment a request is accepted.

QR-code verification adds a physical-handover confirmation step, ensuring that a transaction cannot be marked completed without both the buyer and seller confirming that the exchange actually took place.

The trust system complements this by allowing participants to rate and review each other after a completed exchange, with transaction history, ratings, and reviews combined to calculate and continuously update each student's trust score.

Overall, Module 3 improves the reliability of the Campus Resource Exchange Network (CREN) by ensuring that exchanges are verified and properly recorded. It supports the project's broader goal of creating a trusted academic resource-sharing environment through structured exchange processing, QR verification, and trust scoring.

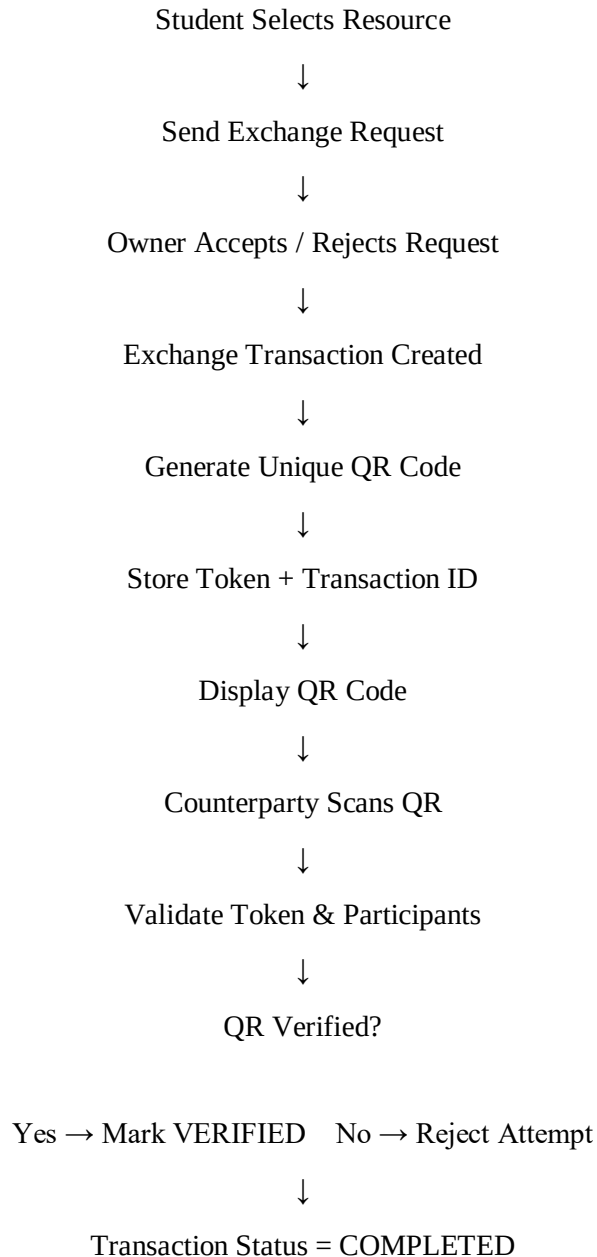
REFERENCES

1. I. Sommerville, Software Engineering, 10th ed., Pearson Education, 2016.
2. R. S. Pressman and B. R. Maxim, Software Engineering: A Practitioner's Approach, 9th ed., McGraw-Hill Education, 2019.
3. OWASP Foundation, OWASP Top 10: The Ten Most Critical Web Application Security Risks, OWASP Foundation.
4. OWASP Foundation, Application Security Verification Standard (ASVS), OWASP Foundation.
5. M. Fowler, Patterns of Enterprise Application Architecture, Addison-Wesley Professional, 2002.
6. ISO/IEC 25010, Systems and Software Engineering – Systems and Software Quality Requirements and Evaluation (SQuaRE).
7. Node.js Documentation, technical documentation for the Node.js runtime environment.
8. Express.js Documentation, technical documentation for REST API and web application development.
9. MySQL Documentation, technical documentation for relational database management, transactions, constraints, and locking.
10. React Documentation, technical documentation for frontend component-based application development.
11. JSON Web Token Documentation, technical reference for token-based authentication.
12. Technical documentation for QR-code generation and QR-code scanning libraries used during ResourceHub implementation.
13. ResourceHub project database schema and API implementation documentation.
14. ResourceHub Module 3 E2E validation and testing documentation.

APPENDICES

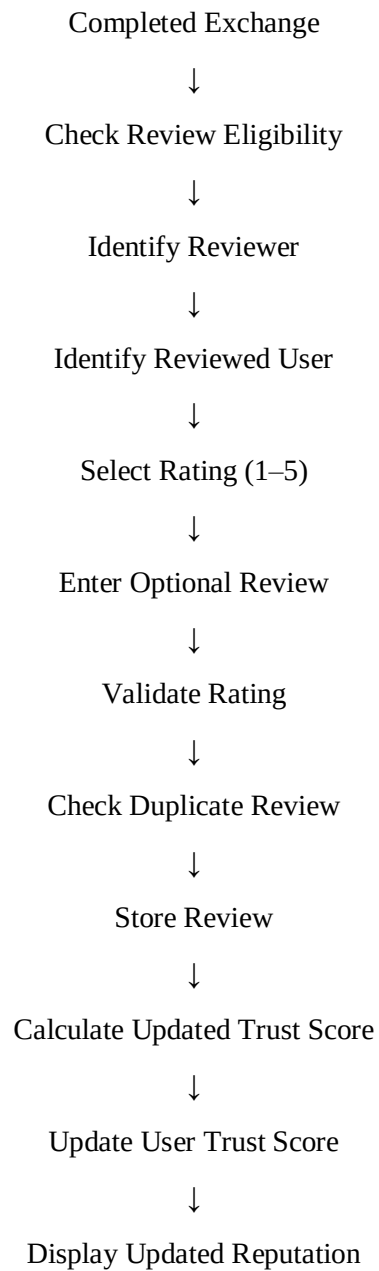
APPENDIX A

EXCHANGE REQUEST & QR HANDOVER WORKFLOW



APPENDIX B

TRUST SYSTEM WORKFLOW



APPENDIX C

EXCHANGE MANAGEMENT IMPLEMENTATION

The exchange-management implementation contains the following major operations:

- 1. Create an exchange request linked to a resource, buyer, and seller.
- 2. Allow the resource owner to accept or reject the request.
- 3. Create an exchange transaction on acceptance.
- 4. Reserve the resource for the duration of the transaction.
- 5. Generate a unique QR code linked to the transaction.
- 6. Validate the QR code and participants during scanning.
- 7. Reject verification attempts from unrelated users.
- 8. Mark the transaction as COMPLETED after successful verification.
- 9. Store the finalised transaction record.

APPENDIX D

TRUST SCORE AND REVIEW IMPLEMENTATION

The review implementation contains the following major operations:

- 1. Validate that the exchange has been completed.
- 2. Identify the authenticated reviewer.
- 3. Identify the participant being reviewed.
- 4. Validate the rating range.
- 5. Accept optional review text.
- 6. Prevent duplicate reviews for the same transaction and reviewer.
- 7. Insert the review record.
- 8. Recalculate the reviewed user's trust score.
- 9. Return the updated trust information to the frontend.

The frontend provides star-rating selection, review text input, submission feedback, and a persistent trust-score display.

APPENDIX E

API ENDPOINTS RELATED TO MODULE 3

API Method	Purpose
POST	Send exchange request
PUT	Accept or reject exchange request
POST	Generate QR code for accepted transaction
POST	Verify QR handover code
POST	Complete verified exchange transaction
POST	Submit rating and review
GET	Retrieve transaction history
GET	Retrieve user trust score

The exact endpoint paths should be copied from the implemented ResourceHub route files before final report submission.

APPENDIX F

DATABASE STRUCTURES RELATED TO MODULE 3

Exchange-management functionality uses the exchange_requests and exchange_transactions tables to maintain:

- Request/Transaction ID
- Buyer and seller identifiers
- Resource ID
- Exchange type (buy/sell/lend/donate/swap)
- Status
- QR verification token
- Created and completed timestamps

The trust and review functionality uses the reviews table to maintain:

- Transaction ID
- Reviewer ID
- Reviewed user ID
- Rating
- Review text
- Creation timestamp

The users table maintains the calculated trust_score associated with each student, and the resources table reflects the current availability status driven by the exchange lifecycle.

APPENDIX G

TESTING CHECKLIST

Test	Expected Outcome
Send exchange request	Request created successfully
Accept exchange request	Transaction created, resource reserved
Reject exchange request	Request closed, resource remains available
Generate QR for transaction	Unique QR code generated
Complete before QR verification	Completion rejected
Verify QR by correct participants	Transaction marked COMPLETED
Unrelated user scans QR	Verification rejected
Submit valid rating	Review stored, trust score updated
Submit duplicate review	Request rejected
Retrieve transaction history	Accurate records returned
Module 3 regression suite	All tests pass